

AN OPERATORIAL GEOMETRY FOR ζ : THE WEIL–TOEPLITZ ALGEBRA AND A LOGISTIC CRITERION FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. We introduce an operatorial geometry for the Weil explicit formula. The renormalized window forms of the explicit formula are organized into a bounded field of finite-dimensional Hilbert spaces over the discrete base $\mathbb{N}$ (an ℓ^∞ -product; no Dixmier continuity is claimed or used); the bounded sections form a C^* -algebra $\mathcal{W}$, the sections vanishing at infinity form a closed ideal $\mathcal{J} \cong c_0$, and the quotient corona $\mathcal{C} = \mathcal{W}/\mathcal{J}$ carries a *symbol* $\sigma(Q)$ whose spectrum is the limit set of the window spectra. The master quantifier (“mean over windows” versus “uniformly over windows”) becomes literally the distinction between the Cesàro states of $\mathcal{C}$ and its norm. Within an axiomatized spectral model $\mathfrak{W}$ we prove a *trichotomy*: off-critical zeros stratify by a resolution-dependent visibility into a fat stratum (seen by the symbol), a thin stratum (accumulating exactly at the boundary point $0 \in \text{spec } \sigma(Q)$), and a sub-resolution stratum (invisible, living in $\mathcal{J}$). The boundary point 0 is exactly the point separating weak from strict essential positivity, and by a sharp characterization theorem it is the unique point on which robust finiteness of the defect depends. Independently, passing to a non-standard normalized signature we prove an exact logistic law $\sigma^{(k)} = \frac{1}{2}(1 - (1 - 2\sigma^0)^{2^k})$ for the tensor tower of window indices, and – in its *localized* form – a criterion $\text{RH} \iff \lim_k \sigma_{\text{loc}}^{(k)} = 0$ whose RH branch ($\text{RH} \Rightarrow 0$) is a theorem and whose $\neg\text{RH}$ branch ($\neg\text{RH} \Rightarrow \frac{1}{2}$) is *conditional* on a single, named, purely analytic gap – a quantitative uniformity in M of Paley–Wiener constants in the witness construction (the Paley–Wiener uniformity problem) – strictly narrower than the realizability hypothesis it replaces. We are explicit about scope: this is a reformulation, not a proof of RH and not progress toward one; the classical wall (uniform second-order pair-correlation asymptotics) is transmuted into the norm of the corona and into a named, non-RH, falsifiable gap; what is genuinely new is the framework – the Weil–Toeplitz algebra – and the reduction of the former realizability hypothesis under $\neg\text{RH}$ to the strictly narrower analytic gap above.

CONTENTS

1. Introduction	3
1.1. The framework in one sentence	3
1.2. What is new	3
1.3. Honest scope, stated up front	3
1.4. Relation to the author’s earlier work	3
Notation	4
2. Defect spaces and the defect δ	4
2.1. Arithmetic of the defect	4
2.2. The exact characterization of robust finiteness	5
3. The Weil–Toeplitz algebra	6
3.1. The field of windows and the corona	6
3.2. The field of Weil	7
4. The spectral model and the trichotomy	7
4.1. The model	7
4.2. The stratification theorems	8
4.3. Synergy of the two axes	10
5. The localized signature and the logistic law	10

5.1. The logistic law	11
5.2. The localized signature	11
5.3. Stability of the cell index	12
5.4. Realizability under $\neg$ RH	13
6. Honest scope	16
6.1. (a) RH is not proved, and no progress toward a proof is claimed	16
6.2. (b) The computability of the symbol is the classical wall, moved	16
6.3. (c) What is genuinely new	17
6.4. What remains open	17
Appendix A. The constructive regime: Weil positivity for finite systems of primes	18
A.1. Pure Euler objects	18
A.2. Additivity: the cross terms vanish exactly	18
A.3. Honest scope	19
Correction history	19
Acknowledgements	20
References	20

1. INTRODUCTION

1.1. The framework in one sentence. The Weil explicit formula defines, on a class of test functions equipped with a notion of height, a hermitian form Q whose positivity is equivalent to the Riemann Hypothesis (RH) [9]. We package the renormalized *windows* of this form as a section of a bounded field of finite-dimensional Hilbert spaces over $\mathbb{N}$, form the associated C^* -algebra $\mathcal{W}$ of bounded sections together with its corona $\mathcal{C} = \mathcal{W}/\mathcal{I}$ (Definition 3.1), and read the obstruction to RH inside the spectrum of the *symbol* $\sigma(Q) \in \mathcal{C}$. This is the exact analogue, for a family, of the passage from a Toeplitz operator to its Calkin image: the essential spectrum is computed window-by-window in the limit.

1.2. What is new. Two things, and they are independent.

(A) *An operatorial geometry.* The corona $\mathcal{C}$ makes the *master quantifier* – the gap between what arithmetic computes in mean and what RH needs uniformly – into a structural feature: membership in c_0 versus ℓ^∞ , equivalently the difference between the Cesàro states of $\mathcal{C}$ and its norm (Section 4, Section 5). Inside this geometry we prove a trichotomy theorem (Theorems 4.5–4.11): off-critical zeros do not split dichotomously into “visible” and “invisible”; they stratify by visibility, and the intermediate stratum lives exactly at the single boundary point $0 \in \text{spec } \sigma(Q)$. That point is, by a sharp characterization (Theorem 2.7), the unique point on which the Fredholm half of the architecture depends.

(B) *A criterion whose RH branch is a theorem.* In normalized non-standard coordinates the tensor tower of window indices obeys an exact logistic recursion (Theorem 5.1), and in its localized version we obtain

$$\text{RH} \iff \lim_{k \rightarrow \infty} \sigma_{\text{loc}}^{(k)} = 0$$

with the RH branch a theorem (compressed Weil positivity, Proposition 5.18) and the $\neg$ RH branch *conditional* on a single, named, purely analytic gap: a quantitative uniformity in M of the Paley–Wiener constants of the witness construction (Hypothesis 5.13, the Paley–Wiener uniformity problem; Theorem 5.16). This gap is strictly narrower than the realizability hypothesis it replaces (Remark 5.17), but it is open: the $\neg$ RH branch is *not* claimed as a theorem here.

1.3. Honest scope, stated up front. We are categorical about this and devote Section 6 to it. (a) We do *not* prove RH and we claim no progress toward a proof: the equivalence of (B) is a reformulation. (b) The “computability of the symbol” – whether $\sigma(Q)$ equals the autonomous arithmetic prediction – is the classical wall (uniform second-order pair-correlation asymptotics) transmuted into the norm topology of the corona; it is not resolved (Section 6, Theorem 6.1). (c) What is genuinely new is the framework and the reduction of the former realizability hypothesis under $\neg$ RH to the strictly narrower analytic gap of the Paley–Wiener uniformity problem (Hypothesis 5.13); that gap is open, and the $\neg$ RH branch is conditional on it. Every result below is labelled: *Theorem* (complete proof here), *Proposition / Lemma* (auxiliary, complete proof here), or *Conjecture / Open / [GAP]* (declared, not proved). When a step rests on a structural input from the author’s earlier work we name it as a cited hypothesis, never assume it tacitly.

1.4. Relation to the author’s earlier work. This paper is self-contained: all proofs are given here. It builds on three earlier works of the author, cited as previous work and never re-proved: the $\kappa = 2m$ bridge identification of the Weil form on a Pontryagin space [10]; the trace criterion and the autonomy of the value of Q computed from primes without zero positions [11]; and the spectral structure of the windowed Weil form (online sea $\approx$ identity, each visible off-critical quadruple a hyperbolic plane) [12]. The precise statements we import are collected as Hypotheses 3.7–3.9 and tabulated in Section 6.

Notation. H denotes a complex separable Hilbert space; $\mathcal{B}(H)$ the bounded operators; $G = G^*$ a bounded self-adjoint operator with spectral measure E_G . A hermitian form is sesquilinear, linear in the first variable, with $Q(y, x) = \overline{Q(x, y)}$. For a subspace W , $\text{neg. ind}(Q|_W)$ is the supremum of $\dim N$ over subspaces $N \subseteq W$ with $Q(x, x) < 0$ for all $x \in N \setminus \{0\}$, valued in $\mathbb{N} \cup \{\infty\}$. A nontrivial zero is written $\rho = \beta + i\gamma$, $\beta \in (0, 1)$; a zero is *off-critical* if $\beta \neq \frac{1}{2}$; a quadruple is $\{\frac{1}{2} \pm \delta \pm i\gamma\}$.

2. DEFECT SPACES AND THE DEFECT δ

We isolate the minimal language of index needed below. The defect of a hermitian form is the dimension of a maximal negative subspace; it is the polarization-theoretic analogue of the Fredholm index.

Definition 2.1 (defect). For a hermitian form Q on a complex vector space H with a marked subspace P , the *defect* is $\delta(Q, P) := \text{neg. ind}(Q|_P) \in \mathbb{N} \cup \{\infty\}$. When H is Hilbert, $P = H$, and $Q(x, y) = \langle Gx, y \rangle$ with $G = G^*$ bounded, we write $\delta(G)$. The form is *polarized* if $\delta = 0$.

Lemma 2.2 (the defect is spectral). $\delta(G) = \dim \text{ran } E_G(-\infty, 0)$.

Proof. On $N_- := \text{ran } E_G(-\infty, 0)$ one has $\langle Gx, x \rangle = \int_{(-\infty, 0)} \lambda d\|E_\lambda x\|^2 < 0$ for $x \neq 0$; thus N_- is negative and $\delta \geq \dim N_-$. Conversely, if N is negative then $E_G(-\infty, 0)|_N$ is injective: if $E_G(-\infty, 0)x = 0$ then $x \in \text{ran } E_G[0, \infty)$, whence $\langle Gx, x \rangle \geq 0$, impossible for $x \in N \setminus \{0\}$. So $\dim N \leq \dim N_-$. $\square$

2.1. Arithmetic of the defect. The defect is monotone, additive, and tensor-multiplicative in a precise sense. These are load-bearing only through the *tensor identity*, which drives the logistic law of Section 5; we re-prove what we use.

Lemma 2.3 (monotonicity and additivity). (i) If $f: (H_1, Q_1, P_1) \rightarrow (H_2, Q_2, P_2)$ is linear, isometric ($Q_2(fx, fy) = Q_1(x, y)$) with $f(P_1) \subseteq P_2$, then $\delta(Q_1, P_1) \leq \delta(Q_2, P_2)$. (ii) $\delta(Q_1 \oplus Q_2) = \delta(Q_1) + \delta(Q_2)$, with no dimension or nondegeneracy hypothesis.

Proof. (i) For $N \subseteq P_1$ negative, $f|_N$ is injective ($fx = 0 \Rightarrow Q_1(x, x) = Q_2(fx, fx) = 0 \Rightarrow x = 0$) and $f(N) \subseteq P_2$ is negative by isometry. (ii) “ $\geq$ ” is direct sum of witnesses. “ $\leq$ ”: if some $\delta(Q_i) = \infty$ there is nothing to prove; if both are finite, choose maximal negatives N_i with $P_i = N_i \oplus R_i$ ($Q_i \geq 0$ on R_i : if $w \in R_i$ had $Q_i(w, w) < 0$ then $N_i \oplus \mathbb{C}w$ would be negative of larger dimension). For $N \subseteq P_1 \oplus P_2$ negative, the projection $\pi: N \rightarrow N_1 \oplus N_2$ along $R_1 \oplus R_2$ has trivial kernel ($\ker \pi = N \cap (R_1 \oplus R_2)$ is both ≥ 0 and < 0), so $\dim N \leq \delta(Q_1) + \delta(Q_2)$. $\square$

Lemma 2.4 (Whitney formula for the defect). For finite-dimensional nondegenerate hermitian forms of signatures (p_1, q_1) , (p_2, q_2) (with $q_i = \delta$),

$$\text{sgn}(Q_1 \otimes Q_2) = (p_1 p_2 + q_1 q_2, p_1 q_2 + q_1 p_2), \quad \delta(Q_1 \otimes Q_2) = p_1 q_2 + q_1 p_2.$$

Proof. Take Q_i -orthonormal bases with signs $\varepsilon_a^{(i)} \in \{\pm 1\}$ (Sylvester’s law of inertia [3]). The tensor basis is $Q_1 \otimes Q_2$ -orthogonal with signs $\varepsilon_a^{(1)} \varepsilon_b^{(2)}$; the sign is -1 exactly when the two factors disagree, giving $p_1 q_2 + q_1 p_2$ negative directions and a diagonal orthogonal basis with no isotropic vectors (nondegeneracy). $\square$

Corollary 2.5 (reduced signature is the only multiplicative character). With $d = p + q$, $s = p - q$: $d(Q_1 \otimes Q_2) = d_1 d_2$ and $s(Q_1 \otimes Q_2) = s_1 s_2$, and $\delta = (d - s)/2$. In particular δ is not multiplicative; the Grothendieck ring of finite nondegenerate forms under $\oplus, \otimes$ is $\mathbb{Z}[\varepsilon]/(\varepsilon^2 - 1)$ via $p + q\varepsilon$, whose only ring homomorphisms to an integral domain are $\varepsilon \mapsto \pm 1$, i.e. d and s .

Proof. $s_1 s_2 = (p_1 - q_1)(p_2 - q_2) = (p_1 p_2 + q_1 q_2) - (p_1 q_2 + q_1 p_2)$ by Lemma 2.4; $d_1 d_2 = (p_1 + q_1)(p_2 + q_2)$ is total dimension. A ring homomorphism to an integral domain sends ε to a root of $\varepsilon^2 = 1$. $\square$

2.2. The exact characterization of robust finiteness. This is the first load-bearing theorem. Essential positivity comes in two strengths, and only the strict one is stable under the compact ideal; the distinction is the whole content.

Definition 2.6 (two essential positivities). $G = G^*$ is *weakly essentially positive* if $\text{spec}_{\text{ess}}(G) \subseteq [0, \infty)$, and *strictly essentially positive* if $\text{spec}_{\text{ess}}(G) \subseteq [c, \infty)$ for some $c > 0$. Since $\text{spec}_{\text{ess}}(G)$ is compact, the latter is equivalent to $\text{spec}_{\text{ess}}(G) \subseteq (0, \infty)$.

Theorem 2.7 (exact characterization of robust finiteness of the defect). *For $G = G^* \in \mathcal{B}(H)$ the following are equivalent:*

- (i) G is strictly essentially positive;
- (ii) $\delta(G + K) < \infty$ for every compact self-adjoint K ;
- (iii) there is $\varepsilon > 0$ with $\delta(G - \varepsilon I) < \infty$.

Proof. (i) $\Rightarrow$ (ii). By Weyl's theorem on stability of the essential spectrum under compact perturbations [5, Thm. IV.5.35], $\text{spec}_{\text{ess}}(G + K) = \text{spec}_{\text{ess}}(G) \subseteq [c, \infty)$. Hence $\text{spec}(G + K) \cap (-\infty, 0]$ is bounded and discrete (its accumulation points would lie in $\text{spec}_{\text{ess}}(G + K)$, disjoint from $(-\infty, 0]$), so finite, of eigenvalues of finite multiplicity, and $\delta(G + K) < \infty$ by Lemma 2.2.

(iii) $\Leftrightarrow$ (i). $\delta(G - \varepsilon I) = \dim \text{ran } E_G(-\infty, \varepsilon)$. If this is finite, $\text{spec}(G) \cap (-\infty, \varepsilon)$ is finitely many eigenvalues of finite multiplicity, so $\text{spec}_{\text{ess}}(G) \subseteq [\varepsilon, \infty)$: strict. Conversely, if $\text{spec}_{\text{ess}}(G) \subseteq [c, \infty)$, take $\varepsilon = c/2$; the same argument gives $\dim \text{ran } E_G(-\infty, c/2) < \infty$.

(ii) $\Rightarrow$ (i), by contraposition. Suppose strictness fails: there is $\mu \in \text{spec}_{\text{ess}}(G)$ with $\mu \leq 0$. We construct a compact self-adjoint K with $\delta(G + K) = \infty$.

Step 1 (orthonormal Weyl sequence with summable error). For each n , $\dim \text{ran } E_G(\mu - 8^{-n}, \mu + 8^{-n}) = \infty$ (this is the defining property of the essential spectrum for self-adjoint operators: if it were finite for some n , μ would be an isolated eigenvalue of finite multiplicity, hence not in spec_{ess} [7, §XIII.4]). Choose inductively $e_n \in \text{ran } E_G(\mu - 8^{-n}, \mu + 8^{-n})$ of unit norm orthogonal to $e_1, \dots, e_{n-1}$ (possible: the range is infinite-dimensional and finitely many orthogonality constraints). Then (e_n) is orthonormal and

$$\|(G - \mu)e_n\|^2 = \int_{(\mu - 8^{-n}, \mu + 8^{-n})} |\lambda - \mu|^2 d\|E_\lambda e_n\|^2 \leq 64^{-n}, \quad \text{i.e.} \quad \|(G - \mu)e_n\| \leq 8^{-n}.$$

Step 2 (Gram matrix of G). For $n \neq m$, by self-adjointness, $|\langle Ge_n, e_m \rangle| = |\langle (G - \mu)e_n, e_m \rangle| \leq 8^{-n}$ and $= |\langle e_n, (G - \mu)e_m \rangle| \leq 8^{-m}$, so $|\langle Ge_n, e_m \rangle| \leq 8^{-\max(n, m)}$; and $\langle Ge_n, e_n \rangle \leq \mu + 8^{-n} \leq 8^{-n}$ (as $\mu \leq 0$).

Step 3 (the perturbation). Set $K := -\sum_{n \geq 1} 2^{-n} \langle \cdot, e_n \rangle e_n$: compact (norm limit of finite-rank), self-adjoint, $\|K\| = \frac{1}{2}$. For $x = \sum_{n \geq n_0} \xi_n e_n$ finitely supported in $\{n \geq n_0\}$,

$$\langle (G + K)x, x \rangle \leq \sum_n (8^{-n} - 2^{-n}) |\xi_n|^2 + \sum_{n \neq m} 8^{-\max(n, m)} |\xi_n| |\xi_m|.$$

The cross term is bounded by $\sum_{n \neq m} 8^{-\max(n, m)} \frac{|\xi_n|^2 + |\xi_m|^2}{2} \leq \sum_n |\xi_n|^2 r_n$ with $r_n = \sum_{m \neq n} 8^{-\max(n, m)} \leq n8^{-n} + \sum_{m > n} 8^{-m} \leq (n+1)8^{-n}$. Hence $\langle (G + K)x, x \rangle \leq \sum_n (8^{-n} + (n+1)8^{-n} - 2^{-n}) |\xi_n|^2$. Since $(n+2)8^{-n} < 2^{-n}$ for all $n \geq 4$, taking $n_0 = 4$ makes every coefficient negative: $G + K$ is negative definite on the infinite-dimensional $V_0 := \overline{\text{span}}\{e_n : n \geq 4\}$, so $\delta(G + K) = \infty$ by Lemma 2.2. $\square$

Remark 2.8 (the design lesson). The only reading of “essential positivity $\Rightarrow$ finite defect” that survives the compact ideal – and the essential spectrum lives modulo that ideal by definition – is the *strict* one. Any certificate must produce a *gap* $c > 0$, not merely a sign. The operator $G = \text{diag}(-1/n)$ has $\text{spec}_{\text{ess}}(G) = \{0\} \subseteq [0, \infty)$ (weakly positive) yet $\delta = \infty$: by Theorem 2.7, this is the generic shape of gaplessness. The (i) $\Rightarrow$ (ii) direction is the form in which earlier work [12] used Weyl's theorem; the contrapositive (ii) $\Rightarrow$ (i), proved above, is what makes the characterization *exact* and is new here.

Lemma 2.9 (gap detector). *If there is an orthonormal sequence (u_n) with $\liminf_n \langle Gu_n, u_n \rangle \leq \ell$, then $\inf \text{spec}_{\text{ess}}(G) \leq \ell$.*

Proof. Suppose $\text{spec}_{\text{ess}}(G) \subseteq [c, \infty)$ with $c > \ell$; fix $\varepsilon \in (0, c - \ell)$ and $E := E_G(-\infty, c - \varepsilon)$, finite-rank by Theorem 2.7(iii) $\Leftrightarrow$ (i). With $m_0 := \inf \text{spec}(G) > -\infty$,

$$\langle Gu, u \rangle \geq m_0 \|Eu\|^2 + (c - \varepsilon) \|(1 - E)u\|^2 = (c - \varepsilon) \|u\|^2 - (c - \varepsilon - m_0) \|Eu\|^2.$$

Since $u_n \rightarrow 0$ and E is finite-rank, $\|Eu_n\| \rightarrow 0$, so $\liminf \langle Gu_n, u_n \rangle \geq c - \varepsilon > \ell$, a contradiction. $\square$

3. THE WEIL–TOEPLITZ ALGEBRA

3.1. The field of windows and the corona.

Definition 3.1 (field of windows; $\mathcal{W}, \mathcal{J}$, the symbol). A *field of windows* is a sequence $\{(H_n, Q_n)\}_{n \geq 2}$ with H_n a finite-dimensional Hilbert space ($\dim H_n = d_n < \infty$), $Q_n = Q_n^* \in \mathcal{B}(H_n)$, and $\sup_n \|Q_n\| < \infty$. Set

$$\mathcal{W} := \left\{ A = (A_n)_n : A_n \in \mathcal{B}(H_n), \|A\| := \sup_n \|A_n\| < \infty \right\}, \quad \mathcal{J} := \{A \in \mathcal{W} : \|A_n\| \rightarrow 0\}.$$

With pointwise operations, $\mathcal{W}$ is a C^* -algebra (the ℓ^∞ product of the field) and $\mathcal{J}$ is a closed two-sided ideal (the *compact ideal of the field*: it is the closure of the finitely supported sections). The *corona* is $\mathcal{C} := \mathcal{W}/\mathcal{J}$ and the *symbol* of a section is its class $\sigma(A) := A + \mathcal{J} \in \mathcal{C}$.

Remark 3.2. In the block-diagonal operator $\bigoplus_n A_n$ on $\bigoplus_n H_n$, a section lies in $\mathcal{J}$ iff the operator is compact; $\mathcal{C}$ is the image of the section algebra in the Calkin algebra of $\bigoplus_n H_n$. The corona is literally “the family modulo compacts”. The name Weil–Toeplitz reflects the analogy with the Toeplitz symbol: the essential spectrum of T_ϕ is the range of the symbol – the “limits of the windows” of the operator [2, 6].

Lemma 3.3 (norm of the symbol). $\|\sigma(A)\|_{\mathcal{C}} = \limsup_n \|A_n\|$.

Proof. ($\leq$) For fixed N the truncation $A^{(N)} := (A_2, \dots, A_N, 0, 0, \dots) \in \mathcal{J}$, so $\|\sigma(A)\| \leq \|A - A^{(N)}\| = \sup_{n > N} \|A_n\| \rightarrow \limsup_n \|A_n\|$. ($\geq$) For every $J \in \mathcal{J}$, $\|A + J\| = \sup_n \|A_n + J_n\| \geq \limsup_n \|A_n + J_n\| = \limsup_n \|A_n\|$ since $\|J_n\| \rightarrow 0$; take the infimum over J . $\square$

Lemma 3.4 (symbol spectrum is the limit set of window spectra). *For $A = (A_n) \in \mathcal{W}$ with each A_n self-adjoint,*

$$\text{spec}_{\mathcal{C}}(\sigma(A)) = \Lambda(A) := \{\lambda \in \mathbb{R} : \exists n_k \rightarrow \infty, \lambda_k \in \text{spec}(A_{n_k}), \lambda_k \rightarrow \lambda\}.$$

Proof. $\sigma(A)$ is self-adjoint, so its spectrum is real; $\Lambda(A)$ is closed and bounded.

($\subseteq$ via contrapositive) If $\sigma(A) - \lambda$ is invertible in $\mathcal{C}$, there is $B \in \mathcal{W}$ with $(A - \lambda)B = 1 + J$, $B(A - \lambda) = 1 + J'$, $J, J' \in \mathcal{J}$. For large n , $\|J_n\|, \|J'_n\| \leq \frac{1}{2}$, so $(A_n - \lambda)B_n$ and $B_n(A_n - \lambda)$ are invertible, whence $A_n - \lambda$ is invertible with $\|(A_n - \lambda)^{-1}\| \leq 2\|B\|$. For self-adjoint A_n , $\|(A_n - \lambda)^{-1}\| = \text{dist}(\lambda, \text{spec } A_n)^{-1}$, so $\text{dist}(\lambda, \text{spec } A_n) \geq (2\|B\|)^{-1}$ for all large n : no $\lambda_k \in \text{spec}(A_{n_k})$ can converge to λ , i.e. $\lambda \notin \Lambda(A)$.

($\supseteq$ via contrapositive) If $\lambda \notin \Lambda(A)$, there are $\varepsilon > 0$ and N with $\text{dist}(\lambda, \text{spec } A_n) \geq \varepsilon$ for all $n \geq N$ (else a violating subsequence would, by boundedness of the spectra, produce a point of Λ within ε of λ). Put $B_n := (A_n - \lambda)^{-1}$ for $n \geq N$ ($\|B_n\| \leq \varepsilon^{-1}$) and $B_n := 0$ for $n < N$; then $B \in \mathcal{W}$ and $(A - \lambda)B - 1$, $B(A - \lambda) - 1$ are finitely supported, hence in $\mathcal{J}$, so $\sigma(B)$ inverts $\sigma(A) - \lambda$. $\square$

Corollary 3.5 (positivity of the symbol is the uniform quantifier). *For $A = A^*$:*

- $\text{spec}(\sigma(A)) \subseteq [0, \infty)$ (weak symbol positivity) $\iff \min \text{spec}(A_n) \geq -\varepsilon_n$ with $\varepsilon_n \rightarrow 0$;
- $\text{spec}(\sigma(A)) \subseteq [c, \infty)$, $c > 0$ (strict symbol positivity) $\iff$ for all $\varepsilon > 0$ there is N with $\text{spec}(A_n) \subseteq (c - \varepsilon, \infty)$ for all $n \geq N$ – a gap uniform over windows, eventually.

Proof. Read Lemma 3.4 twice. $\square$

This is the structural observation that orients the paper: the master quantifier (mean over windows versus uniformly over windows) becomes membership in c_0 versus ℓ^∞ . Corollary 3.5 is the statement “uniform in every window” written as positivity of an element of a C^* -algebra.

3.2. The field of Weil. We now describe the field carrying the Weil form. The structural inputs we import from earlier work are isolated as hypotheses.

Definition 3.6 (the Weil field at a resolution calendar). Fix $\Delta \in (0, 1]$ and a nondecreasing *resolution calendar* $a: [2, \infty) \rightarrow [1, \infty)$ (the log-support budget of the tests per window; the natural choice is $a(T) \asymp \log T$, the resolution scale of individual zeros). For each n : H_n is a finite-dimensional subspace of the test class, concentrated in the window $|\operatorname{Im} s - n| \leq \Delta$ with multiplicative support $\subseteq [e^{-a(n)}, e^{a(n)}]$, endowed with the *online-sea normalization* (the inner product is the spectral Gram form of the online zeros of the window, normalized to mass 1 per zero); and Q_n is the compression of the Weil form to H_n in that normalization. The *autonomous* section $Q^{\text{aut}} = (Q_n^{\text{aut}})$ is the polar/archimedean-plus-near- prime prediction of each window, computable without zero positions [11, 12].

The following are imported, not proved here.

Hypothesis 3.7 (block structure [12]). Each Q_n decomposes, up to controlled couplings, as an online-sea block ($\approx$ identity after normalization) plus one hyperbolic plane $\begin{pmatrix} 0 & \theta \\ \theta & 0 \end{pmatrix}$ for each visible off-critical quadruple in the window.

Hypothesis 3.8 (autonomy of the value [11, 10]). For tests f, g of the class, $Q_n(f, g)$ is computable from the arithmetic blocks of the explicit formula (primes, pole, gamma factor) without any zero position.

Hypothesis 3.9 (spectral realization [10]). The Weil form is realized on a Pontryagin space $(\mathcal{K}, Q)$ with $\kappa(Q) = 2m$, where m is the number of off-critical quadruples; $\text{RH} \iff m = 0$. (*The identification of κ with the negative index of the scaling generator is a conjecture of [10]; we do not use it.*)

Remark 3.10 (honest status of the field). That the compressions exist with the stated normalization, and that the section is bounded, uses the realizability apparatus of [12]. The uniform boundedness $\sup_n \|Q_n\| < \infty$ in the online-sea normalization requires control of the per-window pair-correlation corrections; we treat this as an axiom **(S1)** of the model $\mathfrak{W}$ below (**[GAP]**: per-window pair-correlation boundedness, not proved here), inside which all theorems of Section 4 are proved with complete rigor. This separates, as the contract demands, *theorems* (in the axiomatized model) from the *bridge* (the genuine Weil instance, Section 6).

4. THE SPECTRAL MODEL AND THE TRICHOTOMY

4.1. The model. We axiomatize the spectral structure of the windowed Weil form, distilled from Hypotheses 3.7–3.9.

Definition 4.1 (configuration and visibility). A *configuration* is an at most countable set $Z = \{(\delta_j, \gamma_j)\}_{j \in J}$ with $\delta_j \in (0, \frac{1}{2})$, $\gamma_j \rightarrow \infty$ if $|J| = \infty$, and finitely many j per window; $m := |J|$. ($Z = \emptyset$ is “RH”.) The *visibility* of quadruple j at calendar a is $\theta_j := \theta(\delta_j, \gamma_j) \in [0, 1]$, where θ is a fixed function with

- (V1) θ continuous, nondecreasing in δ , and $\theta(\delta, \gamma) \rightarrow 0$ as $\delta a(\gamma) \rightarrow 0$;
- (V2) there is $c_0 > 0$ with $\theta(\delta, \gamma) \geq c_0$ when $\delta a(\gamma) \geq 1$.

Quadruples with $\delta_j a(\gamma_j) \geq 1$ are *supra-resolution (fat)*; with $\delta_j a(\gamma_j) < 1$, *sub-resolution (thin)*.

Remark 4.2 (origin of θ). (V1)–(V2) are the correct properties, justified by a Paley–Wiener estimate: a test with multiplicative support in $[e^{-a}, e^a]$ has $\hat{f}(s) = \int_{-a}^a F(u) e^{(s-1/2)u} du$, so over a band of width 2δ ,

$$\hat{f}(\tfrac{1}{2} + \delta + i\gamma) \overline{\hat{f}(\tfrac{1}{2} - \delta + i\gamma)} = |\hat{f}(\tfrac{1}{2} + i\gamma)|^2 + O(\delta a e^{a/2} \|F\|_{L^1} \sup_{\text{band}} |\hat{f}|).$$

Thus a quadruple with $\delta a \rightarrow 0$ differs from an online pair by $O(\delta a) \rightarrow 0$ per unit mass (V1); for $\delta a \geq 1$ an aligned test realizes a sign opposition of size $\geq c_0$ (V2), using closure of the class under multiplication by Mellin polynomials [12]. The exact exponent in the transition $\delta a \asymp 1$ is not load-bearing ([**GAP**], declared, non-load-bearing): no theorem below uses more than (V1)–(V2).

Definition 4.3 (the model $\mathfrak{W}(Z)$). Given a configuration Z and calendar a , set

$$H_n := \mathbb{C}^{d_n} \oplus \bigoplus_{j \in J_n^{\text{vis}}} \mathbb{C}^2, \quad Q_n := (I_{d_n} + R_n) \oplus \bigoplus_{j \in J_n^{\text{vis}}} \begin{pmatrix} 0 & \theta_j \\ \theta_j & 0 \end{pmatrix} + C_n,$$

where $d_n \asymp \Delta \log n$, $J_n^{\text{vis}} := \{j : |\gamma_j - n| \leq \Delta, \theta_j > 0\}$, and

- (S1) $\|R_n\| \leq r < \frac{1}{2}$ for all n (sea corrections; [**GAP**] for the Weil instance, Remark 3.10);
- (S2) the hyperbolic blocks carry weight θ_j with (V1)–(V2) (Remark 4.2 for the Weil instance);
- (S3) $\|C_n\| \leq \omega_n$ with $\omega_n \rightarrow 0$ (cross-couplings between separated windows; [**GAP**] for the Weil instance, inherited from correlation decay).

The autonomous section is $Q_n^{\text{aut}} := (I_{d_n} + R_n) \oplus I_{2|J_n^{\text{vis}}|}$. Sub-resolution quadruples with $\theta_j = 0$ do not appear in H_n ; their entire effect is absorbed into R_n/C_n .

All theorems in this section are *unconditional inside $\mathfrak{W}$* ; their transport to ζ carries the statuses of (S1)–(S3) and is discussed in Section 6.

4.2. The stratification theorems. Throughout, $\sigma(Q) \in \mathcal{C}$, $\Lambda(Q) = \text{spec}(\sigma(Q))$, and $\vartheta_n := \max\{\theta_j : j \in J_n^{\text{vis}}\}$ ($:= 0$ if empty).

Lemma 4.4 (per-window spectrum). *For all n , $\text{spec}(Q_n) \subseteq ([1-r, 1+r] \cup \{\pm\theta_j\}_{j \in J_n^{\text{vis}}}) + [-\omega_n, \omega_n]$. In particular $\lambda_{\min}(Q_n) \geq -\vartheta_n - \omega_n$; and if $J_n^{\text{vis}} \neq \emptyset$, $\lambda_{\min}(Q_n) \leq -\vartheta_n + \omega_n$.*

Proof. Without C_n , Q_n is a direct sum: the sea block has spectrum in $[1-r, 1+r]$, and each $\begin{pmatrix} 0 & \theta \\ \theta & 0 \end{pmatrix}$ has eigenvalues $\pm\theta$. The self-adjoint perturbation C_n moves each eigenvalue by at most $\|C_n\| \leq \omega_n$ (finite-dimensional Weyl inequality [1]). The upper bound: the vector $v = (1, -1)/\sqrt{2}$ of the ϑ_n -block gives $\lambda_{\min}(Q_n) \leq \langle Q_n v, v \rangle \leq -\vartheta_n + \omega_n$. $\square$

Theorem 4.5 (fat $\Rightarrow$ the symbol sees them). *If there are $c > 0$ and a subsequence j_k with $\theta_{j_k} \geq c$ and $\gamma_{j_k} \rightarrow \infty$, then $\text{spec}(\sigma(Q)) \cap (-\infty, -c/2] \neq \emptyset$. In particular the symbol is not even weakly positive: the negativity of supra-resolution quadruples survives the quotient by the compact ideal.*

Proof. Let n_k be the window of γ_{j_k} ($n_k \rightarrow \infty$). By Lemma 4.4, $\lambda_{\min}(Q_{n_k}) \leq -c + \omega_{n_k}$. Since $\omega_n \rightarrow 0$ and the spectra are uniformly bounded, a subsequence of $\lambda_{\min}(Q_{n_k})$ converges to a point $\leq -c$, which lies in $\Lambda(Q)$ by Lemma 3.4. $\square$

Theorem 4.6 (thin visible $\Rightarrow$ negatives live exactly at the boundary 0). *Suppose $m = \infty$, $J_n^{\text{vis}} \neq \emptyset$ for infinitely many n , and $\vartheta_n \rightarrow 0$. Then:*

- (i) $0 \in \text{spec}(\sigma(Q))$; if moreover $\vartheta_n > \omega_n$ for infinitely many populated windows, it is attained as a limit of genuinely negative $\lambda_{\min}(Q_n)$ over those windows;
- (ii) $\text{spec}(\sigma(Q)) \subseteq \{0\} \cup [1-r, 1+r]$: weak symbol positivity holds;
- (iii) strict symbol positivity fails.

The negatives accumulate exactly at the single point 0 – the point separating weak from strict positivity, the very point of Theorem 2.7.

Proof. (i) Over the infinitely many populated windows, $\lambda_{\min}(Q_n) \in [-\vartheta_n - \omega_n, -\vartheta_n + \omega_n] \rightarrow 0$ (Lemma 4.4, $\vartheta_n, \omega_n \rightarrow 0$); when $\vartheta_n > \omega_n$ these are genuinely negative, and the limit 0 lies in $\Lambda(Q)$ (Lemma 3.4). (ii) Every eigenvalue of Q_n lies in $[-\vartheta_n - \omega_n, \vartheta_n + \omega_n] \cup [1 - r - \omega_n, 1 + r + \omega_n]$; its limit points lie in $\{0\} \cup [1 - r, 1 + r]$. (iii) Immediate from (i) and Corollary 3.5. $\square$

Theorem 4.7 (strict symbol positivity $\iff$ finite visible count). *Strict symbol positivity at calendar a holds iff $\#\{n : J_n^{\text{vis}} \neq \emptyset\} < \infty$, i.e. $m_a^{\text{vis}} := \#\{j : \theta_j > 0\} < \infty$. (With noise $\omega_n \rightarrow 0$ the equivalence holds verbatim, with no exception clause.) In particular strict positivity says nothing about $m - m_a^{\text{vis}}$, the sub-resolution count.*

Proof. ($\Leftarrow$) If $J_n^{\text{vis}} = \emptyset$ for $n \geq N$, then $\text{spec}(Q_n) \subseteq [1 - r - \omega_n, 1 + r + \omega_n]$ for $n \geq N$, so $\Lambda(Q) \subseteq [1 - r, 1 + r]$ with gap $c = 1 - r > 0$ (Corollary 3.5). ($\Rightarrow$) If infinitely many windows are populated, each carries an eigenvalue $\mu_n \in [-\vartheta_n - \omega_n, -\vartheta_n + \omega_n]$ (Lemma 4.4). The sequence (μ_n) over the populated windows is bounded; any accumulation point ℓ satisfies $\ell \leq \limsup_n (-\vartheta_n + \omega_n) \leq \limsup_n \omega_n = 0$, and $\ell \in \Lambda(Q)$ by Lemma 3.4: strict positivity fails, regardless of the comparison between ϑ_n and ω_n . $\square$

Remark 4.8 (a corrected exception clause). An earlier version of this theorem (and its source, an earlier analysis, Thm. 5.3(ii)) allowed an exception for windows with $\vartheta_n \leq \omega_n$ eventually. That clause is *false* (counterexample: $\omega_n = 1/n$, one population in each even window with $\vartheta_n = 1/(2n)$; the perturbed block eigenvalues lie in $[-3/(2n), 1/(2n)] \rightarrow 0$, so $0 \in \text{spec}(\sigma(Q))$ and there is no gap). The corrected statement above is *stronger*: sub-noise visible windows are *not* compatible with strict symbol positivity. If one wishes to discard populations below the noise floor, the place to do it is the definition of J_n^{vis} (a threshold $\theta_j > \omega$), not an exception in the theorem. Corollary 4.9, Conjecture 4.14 and Theorem 4.13 used only the regime $\theta \geq c_0$ and are unaffected.

Corollary 4.9 (the gap controls the visible count quantitatively). *In the clean case, if $\text{spec}(Q_n) \subseteq [c, \infty)$ for $n \geq N_0$, then $m_a^{\text{vis}} \leq \sum_{n < N_0} |J_n^{\text{vis}}|$, and with the zero-density bound per window $|J_n^{\text{vis}}| \ll \Delta \log n$ [4, Thm. 5.8], $m_a^{\text{vis}} \ll N_0 \log N_0$.*

Proof. By Theorem 4.7 all visible quadruples sit in windows $n < N_0$; sum the per-window count. $\square$

Theorem 4.10 (sub-resolution $\Rightarrow$ the symbol is blind). *Let Z have $m = \infty$ but $\delta_j a(\gamma_j) \rightarrow 0$ (all sub-resolution, with $J_n^{\text{vis}} = \emptyset$). Then, as sections over the same field,*

$$Q - Q^{\text{aut}} \in \mathcal{J}, \quad \text{hence} \quad \sigma(Q) = \sigma(Q^{\text{aut}}) \text{ in } \mathcal{C},$$

and if Q^{aut} has a gap ($\text{spec} \subseteq [1 - r, 1 + r]$ modulo ω), then $\sigma(Q)$ has strict positivity – while the defect of the configuration, as the full form, is $\delta = 2m = \infty$ by Lemma 2.3(ii) applied to the hyperbolic blocks (which exist in the unrestricted test class but require support exceeding the calendar). Strict symbol positivity does not imply $m < \infty$.

Proof. By (V1) in perturbation form (Remark 4.2), the window difference satisfies $\|Q_n - Q_n^{\text{aut}}\| \leq \sum_{j: |\gamma_j - n| \leq \Delta} \varphi(\delta_j a(\gamma_j)) + 2\omega_n$ with $\varphi(t) \rightarrow 0$ as $t \rightarrow 0$. Since $\gamma_j \rightarrow \infty$, finitely many j per window, and $\delta_j a(\gamma_j) \rightarrow 0$, the per-window sum tends to 0: the difference section is in $\mathcal{J}$. The rest is Lemma 3.3 and (S1). The full-form defect is the sum of the hyperbolic planes (Lemma 2.3(ii), each contributing a negative direction in the unrestricted class). $\square$

Theorem 4.11 (calendar no-go: $m < \infty$ is a boundary event over the resolution filtration). (a)

For every calendar a there is a configuration Z with $m = \infty$ and $\delta_j a(\gamma_j) \rightarrow 0$ (take $\gamma_j := j$, $\delta_j := \min(\frac{1}{4}, 1/(j a(\gamma_j)))$): invisible at a by Theorem 4.10.

(b) *For every configuration and quadruple, some calendar sees it ($a(\gamma_j) \geq 1/\delta_j$). Hence with $m_a := \#\{j : \delta_j a(\gamma_j) \geq 1\}$, $m = \sup_a m_a$.*

(c) *Consequently $m < \infty \iff \exists M \forall a : m_a \leq M$, a Σ_2 statement over the filtration of resolutions, and no single Weil–Toeplitz symbol (no fixed calendar) certifies it.*

Proof. (a) $\delta_j a(\gamma_j) \leq 1/j \rightarrow 0$; Theorem 4.10. (b) Monotonicity of $a \mapsto m_a$ and the pointwise observation. (c) Rewriting of (b) together with (a). $\square$

Remark 4.12 (the trichotomy). The dichotomy “symbol or ideal” is false; it is a trichotomy stratified by $\delta_j a(\gamma_j)$:

fat ($\delta_j a(\gamma_j) \geq 1$ i.o.)	in the symbol: $\text{mass} \leq -c/2$	Thm. 4.5
thin visible ($\theta_j \rightarrow 0^+$)	at the boundary $0 \in \text{spec } \sigma(Q)$	Thm. 4.6
sub-resolution ($\delta_j a(\gamma_j) \rightarrow 0$)	in the ideal: $\sigma(Q) = \sigma(Q^{\text{aut}})$	Thm. 4.10

The strata are relative to the calendar (Theorem 4.11); within $\mathfrak{W}$, it is the *intermediate* (*thin-visible*) *stratum* that lives on the symbol/compact boundary, accumulating at 0 – the unique point separating weak from strict positivity, which by Theorem 2.7 is the only point on which robust finiteness of the defect depends – while the fat stratum sits in the negative interior of the symbol and the sub-resolution stratum in the ideal. The slogan “the zeros live on the boundary” is therefore a theorem-in-model for the intermediate stratum only, not a global statement about the zeros of ζ . The master quantifier acquires an exact C^* -algebraic direction: it is the point $0 \in \text{spec } \mathcal{C}$.

4.3. Synergy of the two axes. The Fredholm axis (symbol positivity) and the replication axis combine cleanly.

Theorem 4.13 (replication $\Rightarrow$ the symbol sees). *Suppose Z satisfies the quantitative replication axiom (R): if $m \geq 1$, there is a quadruple $(\delta_0, \gamma_0) \in Z$ and heights $\tau_k \rightarrow \infty$ such that Z contains quadruples $(\delta'_k, \gamma_0 + \tau_k)$ with $\delta'_k \geq \delta_0/2$. Let a be any calendar with $a(\gamma) \rightarrow \infty$. Then $m \geq 1$ forces infinitely many fat quadruples, hence $\text{spec}(\sigma(Q)) \cap (-\infty, -c_0/2] \neq \emptyset$. Contrapositively,*

$$\text{weak symbol positivity} \wedge (R) \implies m = 0 \implies \text{RH}.$$

Proof. The replicas have $\delta'_k a(\gamma_0 + \tau_k) \geq (\delta_0/2)a(\gamma_0 + \tau_k) \rightarrow \infty$, hence are eventually supra-resolution with visibility $\geq c_0$ by (V2). Theorem 4.5 applies. The contrapositive uses only weak positivity since Theorem 4.5 violates even weak positivity. $\square$

The thin/sub-resolution stratum that kills the pure Fredholm route (Theorems 4.10, 4.11) is exactly the stratum that replication does not produce: replicas inherit a fixed δ_0 , and every unbounded calendar eventually sees them. We record the missing piece this leaves open.

Conjecture 4.14 (LP-134: resolution repulsion; new, non-RH, falsifiable). There exist a calendar a (e.g. $a(\gamma) = C \log \gamma$) and $c > 0$ such that every nontrivial zero $\rho = \frac{1}{2} + \delta + i\gamma$ of ζ with $\delta \neq 0$ satisfies $|\delta| \geq c/a(\gamma)$. (“Zeros cannot be asymptotically sub-resolution.”)

Under LP-4.14 the thin and sub-resolution strata are empty, and by Theorems 4.5, 4.7 the Fredholm half becomes a single C^* -statement: $m < \infty \iff$ strict symbol positivity at one calendar. LP-4.14 is *not* RH (a quadruple at $\delta = \frac{1}{4}$ satisfies it and violates RH), follows from nothing known (classical zero-free regions repel from $\text{Re } s = 1$, not from $\text{Re } s = \frac{1}{2}$; density theorems bound how many, not how close), and is of the right shape – a rigidity dichotomy.

5. THE LOCALIZED SIGNATURE AND THE LOGISTIC LAW

We now develop the second, independent contribution: an exact logistic law for the tensor tower of window indices, and an RH criterion in its localized form (one branch a theorem, the other conditional on the Paley–Wiener uniformity problem). We work in non-standard coordinates, using a non-principal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the ultrapower ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathcal{U}$, and the standard part st on the limited elements [8].

5.1. The logistic law.

Theorem 5.1 (exact logistic law of the tensor tower). *Let $\bar{Q}$ be a nondegenerate hermitian form of signature (p, q) on a space of dimension $\bar{n} = p + q$, and set $x_k := \text{neg. ind}(\bar{Q}^{\otimes 2^k})/\bar{n}^{2^k}$. Then for all $k \geq 0$, as an identity of rationals,*

$$x_{k+1} = 2x_k(1 - x_k), \quad 1 - 2x_k = (1 - 2x_0)^{2^k}.$$

Consequently, for any scheme of forms with $\sigma^0 := \text{st}([x_0(N)])$,

$$\sigma^{(k)} := \text{st}([x_k(N)]) = \frac{1}{2}(1 - (1 - 2\sigma^0)^{2^k}), \quad \lim_{k \rightarrow \infty} \sigma^{(k)} = \begin{cases} 0, & \sigma^0 \in \{0, 1\}, \\ \frac{1}{2}, & \sigma^0 \in (0, 1). \end{cases}$$

Proof. By Lemma 2.4 applied to $\bar{Q}^{\otimes 2^{k+1}} = (\bar{Q}^{\otimes 2^k}) \otimes (\bar{Q}^{\otimes 2^k})$, with signature (P_k, K_k) , $P_k + K_k = \bar{n}^{2^k}$,

$$K_{k+1} = 2P_k K_k, \quad P_{k+1} = P_k^2 + K_k^2.$$

Dividing by $\bar{n}^{2^{k+1}} = (\bar{n}^{2^k})^2$ gives $x_{k+1} = 2(P_k/\bar{n}^{2^k})(K_k/\bar{n}^{2^k}) = 2x_k(1 - x_k)$. Then $1 - 2x_{k+1} = 1 - 4x_k + 4x_k^2 = (1 - 2x_k)^2$, so $1 - 2x_k = (1 - 2x_0)^{2^k}$. Each $x_k(N) \in [0, 1]$ is limited; st is a ring homomorphism on limited elements, so the polynomial identity $1 - 2x_{k+1} = (1 - 2x_k)^2$ passes to standard parts for each fixed k (k standard; the identity is used finitely often). Limits: if $\sigma^0 \in (0, 1)$ then $|1 - 2\sigma^0| < 1$ and $(1 - 2\sigma^0)^{2^k} \rightarrow 0$; if $\sigma^0 = 0$, $\sigma^{(k)} \equiv 0$; if $\sigma^0 = 1$, $1 - 2\sigma^0 = -1$ and $(-1)^{2^k} = 1$ for $k \geq 1$, so $\sigma^{(k)} = 0$ – consistent, since the tensor square of a negative-definite form is positive-definite. $\square$

Remark 5.2 (the global signature collapses). The naive global signature – normalizing the window index by the full window dimension $n_N \rightarrow \infty$ – collapses to $\sigma^0 = 0$ under $\neg\text{RH}$ of finite defect, because capturing an off $\times$ on negative plane costs a full dimensional factor $(w_\sigma \log T)(w_\tau \log T)$ while the off slot is rigid ($\leq 4\bar{m}$ vectors): $\kappa/n \lesssim \bar{m}/\max(w \log T) \rightarrow 0$. The same tensor identity that amplifies the tower (Theorem 5.1) dilutes the density of an individual defect. We call this *density invisibility*: a defect of finite cardinal is null for every asymptotic-density functional. This forces the *localized* construction below, where the per-cell dimension is bounded and does not dilute. (The collapse is proved in the analysis preceding this paper, not in [12]; we do not re-prove it, and we do not use the global signature as a criterion.)

5.2. The localized signature.

Definition 5.3 (micro-resolution cell). Fix a *profile* $p = (D, w, M, b, c, c')$ with $D \in \mathbb{N}$ (degree), $w > 0$ (width), $M \in \mathbb{N}$ (decay order), $b \in \mathbb{Q}_{>0}$ (caliber), $(c, c') \in \mathbb{Q}^2$ (centers). The *cell* is

$$V_p := \text{span}\{h_{g_a, \varphi_b} : 0 \leq a, b \leq D\}, \quad \hat{g}_a(\sigma) := B_w^{(M)}(\sigma - ic)\sigma^a, \quad \hat{\varphi}_b(\tau) := B_w^{(M)}(\tau - ic')\tau^b,$$

where $B_w^{(M)}$ is the Mellin transform of a fixed even bump $g_0^{(M, w)} \in C_c^\infty(\mathbb{R}_{>0})$ with

$$|B_w^{(M)}(z)| \leq C_M e^{a_0 |\text{Re } z - 1/2|} (1 + |\text{Im } z|/w)^{-M}, \quad B_w^{(M)}(0) = 1 \quad (*)$$

(Paley–Wiener; $a_0 = a_0(M, w)$ the fixed log-support). Then $\dim V_p \leq (D + 1)^2$, and V_p is test-accessible (the profile mentions no zeros). The *filled cell* is $\tilde{V}_p := V_p \oplus \mathbb{C}^{(D+1)^2}$ with $\tilde{Q}_p := (Q_2|_{V_p}) \oplus I$ (positive filler of dimension equal to the cell’s cap).

Definition 5.4 (localized scheme). For each N , the *grid* $\mathcal{G}_N$ is the finite computable set of profiles p with $D, M \leq N$, $w, b \in \{2^{-j} : j \leq N\}$, and centers in $(b\mathbb{Z})^2 \cap [-2^N, 2^N]^2$. Set

$$x_0(N) := \max_{p \in \mathcal{G}_N} \frac{\text{neg. ind}(\tilde{Q}_p)}{\dim \tilde{V}_p} \in \mathbb{Q} \cap [0, \frac{1}{2}],$$

and $\sigma_{\text{loc}} := \text{st}([x_0(N)])$. For each N fix a winning profile p_N^* (maximizing the ratio), let $\bar{Q}_N$ be the reduced nondegenerate form of $\tilde{Q}_{p_N^*}$ (quotient by the radical) and $\bar{n}_N := \dim \tilde{V}_{p_N^*} - \dim \text{rad}(\tilde{Q}_{p_N^*})$ its dimension, and set

$$x_{\text{base}}(N) := \frac{\text{neg. ind}(\bar{Q}_N)}{\bar{n}_N}, \quad x_k(N) := \frac{\text{neg. ind}(\bar{Q}_N^{\otimes 2^k})}{\bar{n}_N^{2^k}},$$

$$\sigma_{\text{base}}^{\mathcal{U}} := \text{st}([x_{\text{base}}(N)]), \quad \sigma_{\text{loc}}^{(k)} := \text{st}([x_k(N)]).$$

Note that $x_{\text{base}}(N) \neq x_0(N)$ in general when the radical is nontrivial (the denominators differ), and that $\sigma_{\text{base}}^{\mathcal{U}}$ and the intermediate floors $\sigma_{\text{loc}}^{(k)}$, $k \geq 1$, are not claimed to be independent of $\mathcal{U}$: only the limit dichotomy of Proposition 5.6 is.

Proposition 5.5 (well-definedness and autonomy). (a) $x_0(N)$ is determined by the arithmetic data without zero positions: every Gram entry is $Q_2(h, h')$ (autonomy, Hypothesis 3.8), a computable real number (a prime sum), and the grid is finite. Since the entries are computable reals rather than rationals, neg. ind is not claimed to be computed exactly; what holds, and what Section 6 uses, is that the predicate $\text{neg. ind}(\tilde{Q}_p) \geq 1$ is semidecidable by interval arithmetic: a strictly negative eigenvalue is certified at finite precision. (b) $\sigma_{\text{loc}} \in [0, \frac{1}{2}]$ exists for every $\mathcal{U}$. (c) $x_0(N)$ is nondecreasing in N ($\mathcal{G}_N \subseteq \mathcal{G}_{N+1}$), so it converges and $\sigma_{\text{loc}} = \lim_N x_0(N)$ is independent of $\mathcal{U}$.

Proof. (a) The class contains $B_w \cdot$ (monomials) by closure under Mellin multiplication [12]; the filler is an explicit block; a strictly negative eigenvalue of a hermitian matrix with computable-real entries is certified at finite precision by continuity of eigenvalues (Theorem 5.8). (b) st exists on limited elements. (c) Monotone bounded; st of a convergent sequence is its limit, for every $\mathcal{U}$. $\square$

Proposition 5.6 (the logistic law, localized and repaired). For every $\mathcal{U}$ and standard $k \geq 0$,

$$\sigma_{\text{loc}}^{(k)} = \frac{1}{2}(1 - (1 - 2\sigma_{\text{base}}^{\mathcal{U}})^{2^k}), \quad \sigma_{\text{base}}^{\mathcal{U}} \in [\sigma_{\text{loc}}, 2\sigma_{\text{loc}}] \cap [0, \frac{1}{2}].$$

In particular, independently of $\mathcal{U}$: $\lim_k \sigma_{\text{loc}}^{(k)} = 0$ if $\sigma_{\text{loc}} = 0$, and $= \frac{1}{2}$ if $\sigma_{\text{loc}} \in (0, \frac{1}{2}]$.

Proof. For each N , $\bar{Q}_N$ is finite-dimensional nondegenerate; Theorem 5.1 applies coordinatewise with base $x_{\text{base}}(N)$, and st is a ring homomorphism on limited elements, giving the closed formula in $\sigma_{\text{base}}^{\mathcal{U}}$. The bracketing: the filler $I_{(D+1)^2}$ is nondegenerate and orthogonal to the compressed block, so $\text{rad}(\tilde{Q}_p) = \text{rad}(Q_2|_{V_p}) \oplus 0$ and $\bar{n}_N \geq (D+1)^2 = \frac{1}{2} \dim \tilde{V}_{p_N^*}$, whence $\dim \tilde{V}_{p_N^*} / \bar{n}_N \in [1, 2]$; since $\text{neg. ind}(\bar{Q}_N) = \text{neg. ind}(\tilde{Q}_{p_N^*})$ (the radical carries no negative direction) and st is multiplicative on limited elements, $\sigma_{\text{loc}} \leq \sigma_{\text{base}}^{\mathcal{U}} \leq 2\sigma_{\text{loc}}$. Moreover $x_{\text{base}}(N) \leq (\bar{n}_N - (D+1)^2) / \bar{n}_N \leq \frac{1}{2}$, so $\sigma_{\text{base}}^{\mathcal{U}} \in [0, \frac{1}{2}]$ and the case $\sigma^0 = 1$ (the only one breaking the dichotomy in Theorem 5.1) is excluded. The dichotomy: $\sigma_{\text{loc}} = 0 \iff \sigma_{\text{base}}^{\mathcal{U}} = 0$ (bracketing), in which case the tower is identically 0; and $\sigma_{\text{loc}} > 0 \Rightarrow \sigma_{\text{base}}^{\mathcal{U}} \in (0, \frac{1}{2}] \Rightarrow |1 - 2\sigma_{\text{base}}^{\mathcal{U}}| < 1 \Rightarrow \lim_k \sigma_{\text{loc}}^{(k)} = \frac{1}{2}$, for every $\mathcal{U}$. $\square$

Remark 5.7 (corrected). An earlier version of this proposition asserted the closed formula with σ_{loc} itself as the base of the tower. That is *false* as stated whenever the winning Gram is degenerate: the definition then carried two incompatible normalizations of $x_0(N)$ (denominator $\dim \tilde{V}_p$ versus the reduced dimension $\bar{n}_N$). The repair above (found in an adversarial audit) keeps the 0-versus- $\frac{1}{2}$ dichotomy and the $\mathcal{U}$ -independence of the *limit*; the closed formula holds in terms of $\sigma_{\text{base}}^{\mathcal{U}}$, and the intermediate floors are not claimed canonical.

5.3. Stability of the cell index. These three perturbation lemmas make a computable mesh suffice to capture a plane whose exact position is unknown.

Theorem 5.8 (Weyl stability). For hermitian A, E on a finite-dimensional space, with eigenvalues ordered $\lambda_1 \geq \dots \geq \lambda_n$, $|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\|$ for all k . In particular, if A has κ eigenvalues $\leq -\mu$ and $\|E\| < \mu$, then $\text{neg. ind}(A + E) \geq \kappa$.

Proof. Courant–Fischer: $\lambda_k(A + E) = \min_{\text{codim } S=k-1} \max_{0 \neq v \in S} R_{A+E}(v)$ with R the Rayleigh quotient; for all v , $|R_{A+E}(v) - R_A(v)| \leq \|E\|$. Take the optimal S for A and swap $A \leftrightarrow A + E$ [1]. $\square$

Theorem 5.9 (bounded-rank stability). $|\text{neg. ind}(A + E) - \text{neg. ind}(A)| \leq \text{rank } E$.

Proof. Let N be negative for A with $\dim N = \text{neg. ind}(A)$. On $N \cap \ker E$ (codimension $\leq \text{rank } E$ in N), $(A + E)|_{N \cap \ker E} = A|_{N \cap \ker E} < 0$, so $\text{neg. ind}(A + E) \geq \text{neg. ind}(A) - \text{rank } E$. Swap $A \leftrightarrow A + E$ and $E \leftrightarrow -E$. $\square$

Theorem 5.10 (Lipschitz Gram under cell shift). *For a profile p , let $G(\vartheta)$ be the Gram of Q_2 on the cell with centers $(c + \vartheta_1, c' + \vartheta_2)$. Then G is differentiable in ϑ with $\|\partial_{\vartheta_i} G(\vartheta)\| \leq L_p < \infty$, L_p computable from the arithmetic data.*

Proof. Center shift is the modulation $g \mapsto g \cdot x^{i\vartheta}$, whose ϑ -derivative inserts $i(\log x)g$, a test of the same class [12]. The spectral side (1) (below) converges absolutely with local uniformity in ϑ by the decay (*) [12], so one may differentiate under the sum; the derivative inserts a linear factor, i.e. a derived test of the class, whose values are finite and computable. The norm of a $(D+1)^2 \times (D+1)^2$ hermitian matrix is at most its dimension times the maximal entry. $\square$

Theorem 5.11 (the aligned cell survives in the mesh). *Let p^* be an ideal profile (real centers, not necessarily in the mesh) with $G(p^*)$ having κ eigenvalues $\leq -\mu < 0$. Let $p \in \mathcal{G}_N$ be the nearest mesh profile, $|\text{centers}(p) - \text{centers}(p^*)| \leq b\sqrt{2}$. If $b \leq \mu/(4\sqrt{2}L_{p^*})$, then $\text{neg. ind}(G(p)) \geq \kappa$. In particular, for every ideal profile with (μ, L) fixed there is an N_0 , explicit in $(\mu, L, \text{position})$, such that $\mathcal{G}_N$ contains a cell of index $\geq \kappa$ for all $N \geq N_0$. (We say “explicit”, not “computable”: the inputs $(\mu, L, \text{position})$ are not effectively knowable; cf. Section 6(a).)*

Proof. By Theorem 5.10 and the mean value theorem, $\|G(p) - G(p^*)\| \leq L_{p^*} \cdot b\sqrt{2} \cdot 2 = 2\sqrt{2}L_{p^*}b \leq \mu/2 < \mu$; apply Theorem 5.8. Existence of N_0 : the mesh has caliber $2^{-N} \rightarrow 0$ and reach $2^N \rightarrow \infty$. (The earlier bound $b \leq \mu/(2\sqrt{2}L)$ gave only $\leq \mu$, not the strict inequality Theorem 5.8 requires; corrected.) $\square$

5.4. Realizability under $\neg\text{RH}$. We now import the spectral side and prove the realizability theorem. From [12, 10] we have, on the span of the (doubled) test class, the spectral side

$$Q_2(h, h') = \sum_{(\rho_1, \rho_2)} \hat{h}(\rho_1, \rho_2) \overline{\hat{h}'(\iota(\rho_1, \rho_2))}, \quad \iota(\rho_1, \rho_2) := (1 - \bar{\rho}_1, 1 - \bar{\rho}_2), \quad (1)$$

absolutely convergent, with $\hat{h}_{g,\varphi}(\rho_1, \rho_2) = \frac{1}{2}\hat{g}(\sigma)\hat{\varphi}(\tau)$, $\sigma = (\rho_1 + \rho_2)/2$, $\tau = (\rho_1 - \rho_2)/2$; pairs fixed by ι (both online) contribute $|\hat{h}|^2 \geq 0$, and free orbits $\{P, \iota P\}$ contribute hyperbolic planes of signature $(1, 1)$; under RH the sum is diagonal and $Q_2 \geq 0$ [12, 10]. We use $N(t+1) - N(t) \ll \log(2+t)$ [4, Thm. 5.8].

Lemma 5.12 (tail of the concentrated test). *Let $h \in V_p$ with $|\hat{g}(\sigma)| \leq C_g(1 + |\text{Im } \sigma - \gamma_0|)^{-M+D}$ and $|\hat{\varphi}(\tau)| \leq C_\varphi(1 + |\text{Im } \tau|)^{-M+D}$ on $|\text{Re } \tau| \leq 1$. Let Ω_r be the pairs with $\max(|\text{Im } \sigma - \gamma_0|, |\text{Im } \tau|) \geq r$. If $M - D \geq 4$, then*

$$\sum_{(\rho_1, \rho_2) \in \Omega_r} |\hat{h}(\rho_1, \rho_2)| |\hat{h}(\iota(\rho_1, \rho_2))| \leq C_g^2 C_\varphi^2 \frac{C^*(M-D)(\log(2+|\gamma_0|))^2}{(1+r)^{M-D-4}},$$

C^* absolute. Warning (corrected): the constants C_g, C_φ depend on M through the bump (*); the right-hand side tends to 0 as $M \rightarrow \infty$ at fixed r only if the family of bumps keeps $C_g(M)^2 C_\varphi(M)^2 (1+r)^{-(M-D-4)} \rightarrow 0$ – this uniformity is exactly Hypothesis 5.13 below and is *not* proved here.

Proof. Write $u = \operatorname{Im} \sigma = (\gamma_1 + \gamma_2)/2$, $v = \operatorname{Im} \tau = (\gamma_1 - \gamma_2)/2$, so $\gamma_1 = u + v$, $\gamma_2 = u - v$. With $a = |\gamma_1 - \gamma_0|$, $b = |\gamma_2 - \gamma_0|$, the elementary inequality $(1 + \frac{a+b}{2})^2 \geq (1+a)(1+b)$ (since $(1 + \frac{a+b}{2})^2 - (1+a)(1+b) = (\frac{a-b}{2})^2 \geq 0$) and $|u - \gamma_0| + |v| \geq (a+b)/2$ give

$$(1 + |u - \gamma_0|)(1 + |v|) \geq 1 + |u - \gamma_0| + |v| \geq 1 + \frac{a+b}{2} \geq \sqrt{(1+a)(1+b)}.$$

The point $\iota(\rho_1, \rho_2)$ has the same ordinates ($\operatorname{Im}(1 - \bar{\rho}) = \operatorname{Im} \rho$), so with $m := M - D$,

$$|\hat{h}(P)| |\hat{h}(\iota P)| \leq \frac{1}{4} C_g^2 C_\varphi^2 [(1 + |u - \gamma_0|)(1 + |v|)]^{-2m} \leq \frac{1}{4} C_g^2 C_\varphi^2 (1+a)^{-m} (1+b)^{-m}.$$

The double sum factors; the single sum, grouped over unit intervals $|\gamma - \gamma_0| \in [j, j+1)$ with $N(t+1) - N(t) \ll \log(2+t)$, is $\sum_\rho (1 + |\gamma - \gamma_0|)^{-m} \leq C' \zeta(m-1) \log(2 + |\gamma_0|)$, convergent for $m \geq 3$. On Ω_r , one product is $\geq 1+r$; extract $(1+r)^{-(m-4)}$ and retain exponent 2 in each factor for convergence. $\square$

Hypothesis 5.13 (Paley–Wiener uniformity; the Paley–Wiener uniformity problem). There is a family of even bumps $\{g_0^{(M,w)}\}_M$ satisfying (*) such that, in the witness construction of Theorem 5.16 (fixed configuration, fixed D and w), the full tail budget of Lemma 5.12 at $r = 1$ tends to 0 as $M \rightarrow \infty$:

$$C_1(M)^2 C_g(M)^2 C_\varphi(M)^2 C^*(M-D) (\log(2 + |\gamma_0|))^2 2^{-(M-D-4)} \rightarrow 0,$$

where $C_g(M), C_\varphi(M)$ absorb the Paley–Wiener constant C_M and the strip factor $e^{a_0(M,w)}$ of (*), and $C_1(M)$ is the Vandermonde coefficient norm of Lemma 5.15 (which depends on M through $b_0(M) = \min_i |B^{(M)}(z_i - ic)|$).

Remark 5.14 (why Hypothesis 5.13 is a genuine open gap: Paley–Wiener rigidity). For any *fixed* nonzero bump (fixed log-support a_0), the optimal constants C_M in (*) grow superexponentially: for every $A > 1$, $C_M > A^M$ for all large M . Indeed, if $C_M \leq A^M$ for infinitely many M , then for $|t| > Aw$ one would have $|B(t)| \leq (Aw/|t|)^M \rightarrow 0$ along those M , forcing $B \equiv 0$ on $\{|t| > Aw\}$; but B is entire (Paley–Wiener, compact support) and not identically zero, so its real zeros are discrete – contradiction. Hence $C_M^4 2^{-M} \rightarrow \infty$: *no fixed bump satisfies Hypothesis 5.13*, and the standard M -dependent families (powers of sinc, support $a_0(M)$ growing) fail as well – the strip factor $e^{c a_0(M)}$ defeats the polynomial decay in Im at fixed distance $r = 1$. Closing the gap requires either a family of bumps with the stated uniformity, or a different proof mechanism (the classical exponential-dominance route of Weil’s criterion, in which the off-zero term grows like $e^{2\delta_0 a_0}$ with $a_0 \rightarrow \infty$ and absorbs the M -dependent constants). This is *the Paley–Wiener uniformity problem*, open. We emphasize the precise status: the *statement* of Theorem 5.16 is not refuted – Lemma 5.12 is only an upper bound, and the classical mechanism makes the statement plausible – but its proof here is conditional on Hypothesis 5.13.

Lemma 5.15 (Vandermonde with bump). *Let $z_1, \dots, z_q \in \mathbb{C}$ be distinct and B entire with $B(z_i) \neq 0$. If $D \geq q - 1$, the evaluations $f \mapsto f(z_i)$ are linearly independent on $\operatorname{span}\{B(\sigma)\sigma^a : 0 \leq a \leq D\}$; for any data $(y_1, \dots, y_q)$ there is an element with $f(z_i) = y_i$ and coefficient norm $\leq C(z, \min_i |B(z_i)|) \|y\|$. Moreover, for $B = B_w^{(M)}(\cdot - ic) \not\equiv 0$, the centers c with $B(z_i - ic) = 0$ for some i are discrete; every open interval of centers contains a c with $\min_i |B(z_i - ic)| \geq b_0 > 0$, stable on a ball.*

Proof. The matrix $(B(z_i)z_i^a)_{i,a} = \operatorname{diag}(B(z_i)) \cdot (z_i^a)$ has rank q (Vandermonde with distinct nodes, invertible diagonal). The function $c \mapsto B(z_i - ic)$ is entire and not identically zero, so its real zeros are discrete; a finite union of discrete sets is discrete; B is continuous and nonzero on a compact set avoiding the zeros. $\square$

Theorem 5.16 (realizability under $\neg\text{RH}$, conditional on the Paley–Wiener uniformity problem). *Assume $\neg\text{RH}$ and Hypothesis 5.13. Let $\rho^0 = \frac{1}{2} + \delta_0 + i\gamma_0$ be the off-critical zero of minimal height,*

multiplicity $m_0 \geq 1$. Then there are a standard dimension D_0 and a standard N_0 such that for all $N \geq N_0$, $x_0(N) \geq 1/(2D_0) > 0$, and hence for every non-principal $\mathcal{U}$,

$$\sigma_{\text{loc}} \geq \frac{1}{2D_0} > 0, \quad \lim_{k \rightarrow \infty} \sigma_{\text{loc}}^{(k)} = \frac{1}{2}.$$

Proof. Step 0 (target plane). Take $P_0 = (\rho^0, \rho^0)$ with coordinates $(\sigma_0, \tau_0) = (\frac{1}{2} + \delta_0 + i\gamma_0, 0)$ and $\iota P_0 = (1 - \bar{\rho}^0, 1 - \bar{\rho}^0)$ with $(\sigma'_0, 0) = (\frac{1}{2} - \delta_0 + i\gamma_0, 0)$. Since $\delta_0 \neq 0$, $\iota P_0 \neq P_0$: a free orbit, contributing $2m_0^2 \text{Re}[\hat{h}(P_0)\hat{h}(\iota P_0)]$ to (1).

Step 1 (the sets to annihilate). Let S be the finite set of zeros ρ with $||\gamma| - |\gamma_0|| \leq 2$, and Z_σ the finite set of $\sigma = (\rho_1 + \rho_2)/2$ over pairs with both members in S , excluding σ_0, σ'_0 . Let Z_τ be the finite set of $\tau = (\rho_1 - \rho_2)/2$ over the *resonant pairs*: pairs with both members in S and $\sigma \in \{\sigma_0, \sigma'_0\}$, other than the target orbit $\{P_0, \iota P_0\}$. Every resonant pair has $\tau \neq 0$ ($\tau = 0$ together with $\sigma = \sigma_0$ forces $\rho_1 = \rho_2 = \rho^0$, the target). Put $D := \max(|Z_\sigma| + 1, |Z_\tau|)$; all standard, fixed by the configuration. (The annihilation of the resonant pairs is a correction: an earlier version of this proof omitted them, and under general $\neg$ RH configurations they contribute terms of uncontrolled sign.)

Step 2 (the cell and witness). Choose the ideal profile p^* with degree D , σ -center c^* where Lemma 5.15 gives $\min |B(z - ic^*)| \geq b_0$ over $Z_\sigma \cup \{\sigma_0, \sigma'_0\}$, and τ -center c'^* where it gives the same over $Z_\tau \cup \{0\}$. In V_{p^*} build $v = h_{g_v, \varphi_v}$ with $\hat{g}_v = 0$ on Z_σ , $\hat{g}_v(\sigma_0) = 1$, $\hat{g}_v(\sigma'_0) = -1$, and $\hat{\varphi}_v = 0$ on Z_τ , $\hat{\varphi}_v(0) = 1$ (Lemma 5.15 on each side; coefficient norm $\leq C_1$, standard at fixed M but M -dependent, cf. Hypothesis 5.13). The data $(1, -1)$ fixes the phase: the target plane contributes $2m_0^2 \text{Re}[(\frac{1}{2} \cdot 1 \cdot 1)(\frac{1}{2} \cdot (-1) \cdot 1)] = -m_0^2/2 < 0$.

Step 3 (budget of $Q_2(v, v)$). Decompose (1): (a) the orbit $\{P_0, \iota P_0\}$ contributes $-m_0^2/2$; (b) pairs with both members in S other than the target: either $\sigma \in Z_\sigma$, where $\hat{g}_v = 0$, or $\sigma \in \{\sigma_0, \sigma'_0\}$ (resonant), where $\tau \in Z_\tau$ and $\hat{\varphi}_v = 0$ – in both cases the contribution is 0 (the case split is exhaustive, and the ι -image adds no new case: $\sigma(\iota P) = 1 - \bar{\sigma}$ swaps $\sigma_0 \leftrightarrow \sigma'_0$, already covered); (c) the rest lie in Ω_1 about $(\gamma_0, 0)$ (some member outside S), and by Lemma 5.12 with $r = 1$ the bound is the budget of Hypothesis 5.13, which *by that hypothesis – and only by it (Remark 5.14); this is GAP-A, the conditional point of the theorem* – tends to 0 as $M \rightarrow \infty$. Choose $M = M_0$ standard with $|c| \leq m_0^2/4$. Then

$$Q_2(v, v) \leq -\frac{m_0^2}{2} + 0 + \frac{m_0^2}{4} = -\frac{m_0^2}{4} < 0,$$

so $\text{neg. ind}(Q_2|_{V_{p^*}}) \geq 1$, with a standard spectral margin $-\mu_0$.

Step 4 (ideal cell to grid). By Theorem 5.11 (with $\kappa = 1$, $\mu = \mu_0$, $L = L_{p^*}$ finite by Theorem 5.10, all standard) there is a standard N_0 such that for $N \geq N_0$, $\mathcal{G}_N$ contains a profile p with $\text{neg. ind}(Q_2|_{V_p}) \geq 1$.

Step 5 (conclusion). For $N \geq N_0$, $x_0(N) \geq \text{neg. ind}(\tilde{Q}_p)/\dim \tilde{V}_p \geq 1/(2(D+1)^2) =: 1/(2D_0)$ with $D_0 = (D+1)^2$ standard (filler doubles dimension; the negative subspace persists). By Proposition 5.5(c), $\sigma_{\text{loc}} = \lim x_0(N) \geq 1/(2D_0)$ for every $\mathcal{U}$; Proposition 5.6 gives $\sigma_{\text{loc}}^{(k)} \rightarrow \frac{1}{2}$. $\square$

Remark 5.17 (the conceptual key; what replaces the realizability hypothesis). The index $\text{neg. ind}(Q_2|_{V_p})$ is a property of the *compressed Gram*. Test-accessibility restricts the *specification* of V_p (no zero positions); the *witness* v inside V_p is an object of the proof, free to use all positions. Earlier work [12] used polynomial factors with roots at zero positions and so appeared to inherit a realizability caveat (the ‘‘Hypothesis D’’ of [12]); here the same factors are rebuilt as linear combinations *inside* the test-accessible span $\{B \cdot \sigma^a\}$ (Lemma 5.15). Localizing to the bounded height of the defect dissolves the $T \rightarrow \infty$ regime of that caveat – but, as the adversarial audit established, it does not dissolve the caveat entirely: it *transmutes* it into the uniformity in M of the Paley–Wiener constant budget, Hypothesis 5.13 (the Paley–Wiener uniformity problem), a strictly narrower and purely analytic statement, open. *We claim no quantitative bound:* D_0, N_0 are not effective in $\bar{m}$ – they

depend on the unknown local configuration. That non-effectivity is exactly what the criterion does not need.

Proposition 5.18 (RH branch). *Under RH, $x_0(N) = 0$ for all N , $\sigma_{\text{loc}} = 0$, and $\sigma_{\text{loc}}^{(k)} \equiv 0$.*

Proof. Under RH the spectral side (1) is diagonal, so $Q_2 \geq 0$ on the span [12, 10]; every compression of a positive form is positive; the filler is positive; $\text{neg. ind} = 0$ in every cell. $\square$

Theorem 5.19 (the logistic criterion). *For every non-principal ultrafilter $\mathcal{U}$:*

- (i) (theorem) $\text{RH} \implies \sigma_{\text{loc}} = 0 \implies \lim_k \sigma_{\text{loc}}^{(k)} = 0$;
- (ii) (conditional on Hypothesis 5.13, the Paley–Wiener uniformity problem) $\neg\text{RH} \implies \sigma_{\text{loc}} > 0 \implies \lim_k \sigma_{\text{loc}}^{(k)} = \frac{1}{2}$.

Hence, conditionally on Hypothesis 5.13,

$$\text{RH} \iff \sigma_{\text{loc}} = 0 \iff \lim_{k \rightarrow \infty} \sigma_{\text{loc}}^{(k)} = 0 \iff \lim_k \sigma_{\text{loc}}^{(k)} \neq \frac{1}{2}.$$

Unconditionally, $\lim_k \sigma_{\text{loc}}^{(k)} \in \{0, \frac{1}{2}\}$ with $0 \iff \sigma_{\text{loc}} = 0$, independently of $\mathcal{U}$.

Proof. (i) Proposition 5.18 and Proposition 5.6. (ii) Theorem 5.16 and Proposition 5.6. The $\mathcal{U}$ -independence of σ_{loc} is Proposition 5.5(c); the unconditional dichotomy is Proposition 5.6. $\square$

Remark 5.20 ($\mathcal{U}$ is tamed for free). The dichotomy of Theorem 5.19 is independent of the ultrafilter: σ_{loc} is a genuine limit (Proposition 5.5(c)). The non-canonicity one might fear – “which ultrafilter concentrates where the defect lives?” – is resolved by monotonicity of the growing grid, not by a choice.

6. HONEST SCOPE

We state precisely what is and is not achieved, per the contract.

6.1. (a) RH is not proved, and no progress toward a proof is claimed. Theorem 5.19 is an *equivalence conditional on Hypothesis 5.13* (its RH branch is an unconditional theorem), a reformulation of RH. It does not decide RH and provides no new route to deciding it. The standard part $\sigma_{\text{loc}} = 0$ is a Π_1 statement over the sequence $(x_0(N))_N$ – “for all large N , $x_0(N)$ small” – exactly as non-effective as RH itself. The improvement is one of *coordinates*: each event $x_0(N)$ is now finite linear algebra over arithmetic data (Proposition 5.5(a)), so falsifying RH becomes “certify, by interval arithmetic, a negative eigenvalue of an arithmetic Gram” rather than “integrate ζ'/ζ with controlled error”. This is a second semidecision machine alongside the classical numerical one – with the caveat that its *completeness* (that $\neg\text{RH}$ forces the machine to halt) is Theorem 5.16, hence conditional on the Paley–Wiener uniformity problem; RH gains nothing.

6.2. (b) The computability of the symbol is the classical wall, moved. The trichotomy (Section 4) locates the obstruction at the single point $0 \in \text{spec } \sigma(Q)$, and the following identity shows that deciding whether the symbol equals its autonomous prediction is the classical wall in new coordinates.

Theorem 6.1 (computing the symbol = uniformity over windows). *On the same field of test spaces, $\|\sigma(Q) - \sigma(Q^{\text{aut}})\|_{\mathcal{C}} = \limsup_n \|Q_n - Q_n^{\text{aut}}\|$. Hence the statement “the symbol of the Weil form is the archimedean/near-prime prediction” is equivalent to $\|Q_n - Q_n^{\text{aut}}\| \rightarrow 0$: the second-order deficit asymptotics, in operator norm per window, uniformly over n on the whole ball of tests.*

Proof. Lemma 3.3 applied to the difference section; the right-hand side is the per-window deficit form against the window’s tests [12]. $\square$

The right-hand side is the uniform second-order pair-correlation asymptotic – the classical wall. What arithmetic computes unconditionally are the *Cesàro states* (the mean over windows) of $\sigma(Q)$; strict positivity is a statement of *norm* (all states, including the pure states supported on window subsequences aligned with the zeros). The gap between the computable and the necessary is exactly the gap between the mean states and the exceptional pure states of the corona. We record the abstract half of what is provable.

Proposition 6.2 (mean Cesàro positivity, abstract half). *If, for every $\varepsilon > 0$, $N^{-1}\#\{n \leq N : \lambda_{\min}(Q_n) < -\varepsilon\} \rightarrow 0$ (“almost every window is ε -positive”), then for every positive test section $0 \leq A_n \leq I$ and every Banach limit $\mathcal{L}$,*

$$\mathcal{L}\text{-}\lim_N \frac{1}{N} \sum_{n \leq N} \frac{\text{tr}(Q_n A_n)}{d_n} \geq 0 :$$

all positive Cesàro states of $\sigma(Q)$ are ≥ 0 .

Proof. If $\lambda_{\min}(Q_n) \geq -\varepsilon$, then $Q_n \geq -\varepsilon I$, so $\text{tr}(Q_n A_n) \geq -\varepsilon \text{tr}(A_n) \geq -\varepsilon d_n$ (using $\text{tr}(CA) \geq 0$ for $C, A \geq 0$); otherwise the trivial bound $\text{tr}(Q_n A_n) \geq -\sup_k \|Q_k\| d_n$. The bad windows have vanishing density, so the Banach limit of a sequence $\geq -\varepsilon - o(1)$ is $\geq -\varepsilon$; let $\varepsilon \rightarrow 0$. $\square$

The hypothesis is the mean-value / large-sieve output (almost every window), whose arithmetic status is inherited from the author’s earlier work and is *not* proved here ([**GAP**], declared); Proposition 6.2 guarantees that once supplied, it yields “mean positivity of the symbol”, delimiting the wall from inside: the wall is precisely the passage from mean states to pure states.

6.3. (c) What is genuinely new.

- The *framework*: the Weil–Toeplitz algebra $\mathcal{W}$, its corona $\mathcal{C}$, and the symbol $\sigma(Q)$ (Definition 3.1, Lemmas 3.3–3.4), which realize the master quantifier as c_0 versus ℓ^∞ (Corollary 3.5).
- The *exact* characterization of robust defect finiteness (Theorem 2.7, the (ii) $\Rightarrow$ (i) direction), pinning the obstruction to the single boundary point 0.
- The *trichotomy* (Theorems 4.5–4.11): off-critical zeros stratify by visibility – the thin-visible stratum on the symbol/compact boundary, the fat stratum in the symbol, the sub-resolution stratum in the ideal (Remark 4.12) – and $m < \infty$ is a Σ_2 boundary event over the resolution filtration, certified by no fixed calendar.
- That the *realizability hypothesis* of [12] is replaced by the strictly narrower, purely analytic gap of Hypothesis 5.13 (the Paley–Wiener uniformity problem): test-accessibility is no longer the obstruction (Remark 5.17); what remains is a constant budget, open.
- The exact logistic law (Theorem 5.1) and the criterion (Theorem 5.19), whose RH branch is a theorem and whose $\neg$ RH branch is conditional on the Paley–Wiener uniformity problem.

6.4. What remains open. The computability of the symbol – uniform pair-correlation asymptotics (Theorem 6.1) – is the old wall, intact. the Paley–Wiener uniformity problem (the Paley–Wiener constant budget, Hypothesis 5.13 and Remark 5.14) is the single analytic gap of the $\neg$ RH branch of Theorem 5.19. LP-4.14 (resolution repulsion) is a new, non-RH, falsifiable lemma; together with replication (R) it would close the Fredholm half. The non-effectivity is Π_1 , pure (Section 6(a)). We make the dependence on cited inputs explicit.

Remark 6.3 (cited hypotheses from earlier work on which the paper depends). Theorem 5.19, the central equivalence, depends on:

- (1) the spectral side (1) with absolute convergence and the factorization $\hat{h}_{g,\varphi} = \frac{1}{2}\hat{g}(\sigma)\hat{\varphi}(\tau)$, and the free-orbit / hyperbolic-plane structure with multiplicity weights, from [12] (and [10] for the realization $(\mathcal{K}, Q)$, $\kappa = 2m$);
- (2) the autonomy of the value: $Q_2(h, h')$ computable from primes without zero positions, from [11, 10];

- (3) closure of the test class under multiplication by Mellin polynomials, from [12];
- (4) under RH, diagonality of (1) (hence $Q_2 \geq 0$), from [12];
- (5) the zero-density bound $N(t+1) - N(t) \ll \log(2+t)$ [4, Thm. 5.8] (published literature, not P-series);
- (6) for the $\neg$ RH branch only: Hypothesis 5.13 (the Paley–Wiener uniformity problem), *open* – not a cited input but a declared gap.

The trichotomy (Section 4) additionally depends on the model axioms (S1)–(S3) of Definition 4.3, of which: the block structure and the sub-resolution blindness (V1) are theorems of [12]; (S1) (per-window pair-correlation boundedness) and (S3) (cross-window decay) are declared [GAP]s for the genuine Weil instance; (V2) carries the status of the realizability lemma of [12]. Theorem 5.16 uses items (1)–(3), (5) and the open Hypothesis 5.13 (item (6), the Paley–Wiener uniformity problem), but *not* (S1)–(S3) and *not* the realizability lemma of [12] (Remark 5.17). Proposition 6.2 is unconditional given its stated density hypothesis, whose arithmetic provenance (mean-value / large sieve) is a declared [GAP].

APPENDIX A. THE CONSTRUCTIVE REGIME: WEIL POSITIVITY FOR FINITE SYSTEMS OF PRIMES

The Weil–Toeplitz analysis of the body concerns the *defect* — the off-line zeros and the negative directions they create. It is illuminating to record, by way of contrast, the opposite extreme: the fully constructive regime, in which positivity is immediate and the whole content lies in a single decidable discriminator. The material of this appendix is the easy direction of Weil’s criterion [9], recast categorically; we claim no novelty for the positivity itself and are explicit about its standing.

A.1. Pure Euler objects. For a prime p write $L_p = \log p$. The *Euler object* of p with parameter α ($1 \leq |\alpha| \leq p$) is the datum $F_{p,\alpha}$ whose local zeta factor is $(1 - \alpha p^{-s})^{-1}(1 - (p/\bar{\alpha})p^{-s})^{-1}$, with Weil functional $W_{F_{p,\alpha}}$ assembled from the coefficients $c_m = \alpha^m + (p/\bar{\alpha})^m$ of its logarithmic derivative. The object is *self-dual* (invariant under $s \mapsto 1-s$), and we say it satisfies the positivity property H if its localized Weil quadratic form $Q_{F_{p,\alpha}}(f, f) = W_{F_{p,\alpha}}(f \star \bar{f}) \geq 0$ for all admissible f .

Theorem A.1 (Local purity is the exact discriminator). *For $1 \leq |\alpha| \leq p$,*

$$F_{p,\alpha} \text{ satisfies } H \iff |\alpha| = \sqrt{p} \iff |c_m| \leq 2p^{m/2} \quad \forall m \geq 1.$$

Proof. If $|\alpha| = \sqrt{p}$ then $p/\bar{\alpha} = \alpha$, the two geometric progressions coincide on $\Re s = \frac{1}{2}$, and $c_m = 2\alpha^m$ with $|c_m| = 2p^{m/2}$. Summing the single-factor Poisson identity over the doubled progression,

$$Q_{F_{p,\alpha}}(f, f) = \sum_{\rho} \widehat{f}(\rho) \overline{\widehat{f}(\frac{1}{2} + i\Im \rho)} = 2 \sum_{\xi \in \Lambda_p} |\widehat{F}(\xi)|^2 \geq 0,$$

since every zero ρ has $\Re \rho = \frac{1}{2}$ (so $1 - \bar{\rho} = \rho$ and each term is a squared modulus), where $\Lambda_p = \{(\arg \alpha + 2\pi k)/L_p : k \in \mathbb{Z}\}$. Conversely, if $|\alpha| = p^{1/2+\delta'}$ with $\delta' \neq 0$, the reverse triangle inequality gives $|c_m| \geq p^{m/2}(p^{m\delta'} - p^{-m\delta'}) > 2p^{m/2}$ for large m , the zeros leave the line, and H fails. Equality $|\alpha| = \sqrt{p}$ is exactly $|c_m| = 2p^{m/2}$. $\square$

A.2. Additivity: the cross terms vanish exactly.

Proposition A.2 (Convolution does not mix Weil data). *For Euler objects X_1, X_2 of distinct primes, the Weil functional of the product is additive, $W_{X_1 \star X_2} = W_{X_1} + W_{X_2}$, with cross terms exactly zero. Consequently a product of pure Euler objects over any finite set of primes satisfies H if and only if each factor does, i.e. if and only if each is pure ($|\alpha_p| = \sqrt{p}$).*

Proof. The logarithmic derivative of a product is the sum of the logarithmic derivatives, and the Weil functional is linear in it; the cross terms are integrals of $e^{i(\log p)\xi} \bar{e}^{i(\log q)\xi}$ against the test data, which vanish because $\log p \neq \log q$ places the two progressions on disjoint frequency lattices. Hence positivity is per-prime, and Theorem A.1 applies factorwise. $\square$

A.3. Honest scope. Proposition A.2 with Theorem A.1 is exactly the *easy* direction of Weil’s criterion: purity places every zero on $\Re s = \frac{1}{2}$ *by hypothesis*, and positivity follows term by term, with the additivity of logarithmic derivatives (classical since Euler) making the finite system no harder than one prime. We claim no novelty for the positivity. Two structural points are worth recording. First, purity is the *exact, decidable* discriminator: H holds if and only if $|\alpha_p| = \sqrt{p}$ for every p in the system, a sharp local condition. Second, the cross-prime interference, which one might fear, is located precisely: it is absent from the *sign* (the cross terms are identically zero) and present only in the spectral structure that a genuine global object would carry. The finite Euler product is *not* ζ : the constructive positivity survives every finite truncation but the spectrality does not, and ζ is not the limit of its Euler factors in the relevant sense. The contrast with the body is the point: in the constructive regime purity is assumed and positivity is free, whereas the difficulty addressed in the main text is precisely the absence of an a priori purity.

CORRECTION HISTORY

June 2026. Two adversarial audits mandated the following corrections, all applied in this version. A false victory being worse than a failure, we record them explicitly.

- **(Critical.)** The $\neg$ RH branch of the logistic criterion was previously claimed as a theorem over cited inputs. Its proof contained a constant treated as constant while depending on the parameter sent to infinity: the tail budget of Theorem 5.16, Step 3, used M -dependent Paley–Wiener constants whose uniformity is not only unproved but *false* for every fixed bump (Paley–Wiener rigidity, Remark 5.14). Status downgraded: the branch is now *conditional* on Hypothesis 5.13 (the Paley–Wiener uniformity problem). Abstract, introduction, Theorem 5.19 and Section 6 reworded accordingly. The statement itself is not refuted; the RH branch (Proposition 5.18) and the exact logistic algebra (Theorem 5.1) survive the audit intact.
- **(Correction.)** The same proof failed to annihilate the *resonant pairs* (both members in S , $\sigma \in \{\sigma_0, \sigma'_0\}$, off the target orbit), which exist under general $\neg$ RH configurations and contribute terms of uncontrolled sign. Repaired here (repair validated by the audit) by Vandermonde annihilation on the τ side over Z_τ , enlarging D ; the enlargement feeds back into the constant budget, so the two gaps must be closed jointly within the Paley–Wiener uniformity problem.
- **(Correction.)** The closed logistic formula in σ_{loc} (the former version of Proposition 5.6) was false when the winning Gram is degenerate (two incompatible normalizations of $x_0(N)$: the radical bug). Repaired via $\sigma_{\text{base}}^{\mathcal{U}} \in [\sigma_{\text{loc}}, 2\sigma_{\text{loc}}] \cap [0, \frac{1}{2}]$ (Remark 5.7); the 0-versus- $\frac{1}{2}$ dichotomy and the $\mathcal{U}$ -independence of the limit survive; intermediate floors are no longer claimed canonical.
- **(adversarial audit)** “neg. ind computed exactly” weakened to semidecidability by interval arithmetic (Proposition 5.5); mesh constant corrected to $b \leq \mu/(4\sqrt{2}L)$ and “computable N_0 ” to “explicit N_0 ” (Theorem 5.11); the unconditional “ $\rightarrow 0$ as $M \rightarrow \infty$ ” claim removed from Lemma 5.12; the attribution of the global-signature collapse corrected (Remark 5.2).
- **(adversarial audit)** The noisy-case exception clause of Theorem 4.7 (“up to windows with $\vartheta_n \leq \omega_n$ eventually”) was *false* (counterexample: $\omega_n = 1/n$, $\vartheta_n = 1/(2n)$). The correction *strengthens* the theorem: strict symbol positivity $\iff$ finitely many visible windows, with no exception (Remark 4.8). Downstream results unaffected.
- **(adversarial audit)** “Continuous field” replaced by “bounded field over the discrete base $\mathbb{N}$ ” (over a discrete base the Dixmier continuity condition is vacuous; no proof used it); the global slogan of Remark 4.12 stratified (only the thin-visible stratum lives on the symbol/compact boundary); a typographic inequality chain in Lemma 4.4 fixed; Theorem 4.6(i) qualified by $\vartheta_n > \omega_n$ i.o. for the negativity claim. Theorem 2.7 and Lemmas 3.3–3.4 passed the audit without correction.

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